

Python – One Month Teaching Schedule

Student: 2nd Year Mechanical Engineering **Mode:** Online

Device: Mobile only **Class Time:** 6:00 PM – 7:00 PM **Start:** Friday

Course Goal: Python basics → problem solving → practical engineering examples → mini project.

Week 1 – Python Basics

Day	Topic	What to Teach
Day 1 – Friday	Introduction + First Program	Python introduction, engineering use cases, mobile setup, print(), comments, basic syntax
Day 2 – Monday	Variables & Data Types	Variables, int, float, str, bool, type()
Day 3 – Tuesday	Input & Output	input(), type conversion, user-input programs
Day 4 – Wednesday	Operators	Arithmetic, comparison, logical, assignment operators
Day 5 – Thursday	Practice Day	Calculator, rectangle area, circle area, temperature conversion

Week 2 – Decision Making + Loops

Day	Topic	What to Teach
Day 6 – Friday	if	Conditions, indentation, simple decision programs
Day 7 – Monday	if-else	Even/odd, positive/negative, pass/fail
Day 8 – Tuesday	elif	Grade calculator, largest number
Day 9 – Wednesday	for Loop	range(), counting, multiplication tables
Day 10 – Thursday	while Loop	Repeated execution, counters, simple loops

Week 3 – Strings, Collections & Functions

Day	Topic	What to Teach
Day 11 – Friday	Strings	Indexing, slicing, common string methods
Day 12 – Monday	Lists	Indexing, append(), remove(), sort(), list practice
Day 13 – Tuesday	Tuple & Set	Tuple basics, sets, duplicate removal
Day 14 – Wednesday	Dictionary	Key-value pairs, add/update/delete data
Day 15 – Thursday	Functions	def, parameters, arguments, return

Week 4 – Practical Python + Mini Project

Day	Topic	What to Teach
Day 16 – Friday	File Handling	open(), read(), write(), basic file handling
Day 17 – Monday	Error Handling	try, except, common errors
Day 18 – Tuesday	Modules	import, math and useful built-in modules

Day	Topic	What to Teach
Day 19 – Wednesday	Problem Solving	5–10 coding problems and logic building
Day 20 – Thursday	Mini Project	Engineering Calculator – build, test and explain

Daily 1-Hour Class Structure

Time	Activity
6:00 – 6:10	Previous class revision
6:10 – 6:30	New concept explanation
6:30 – 6:50	Live coding + student typing practice
6:50 – 7:00	Student practice + homework

Mechanical Engineering Practice Ideas

Rectangle/Circle area • Speed • Distance • Temperature conversion • Force calculation • Simple engineering formulas

Mini Project – Engineering Calculator

Student chooses an option from a menu:

1. Rectangle Area
2. Circle Area
3. Speed
4. Distance
5. Force
6. Temperature Conversion
7. Exit

This project reinforces input, variables, operators, if/elif, loops, functions and modules.

Day 1 Detailed Flow

6:00 – 6:10: Introduction
6:10 – 6:20: What is Python? Where is it used in engineering?
6:20 – 6:30: Mobile-based Python setup
6:30 – 6:40: print() and basic syntax
6:40 – 6:50: Student types first programs
6:50 – 6:55: 3 small practice programs
6:55 – 7:00: Homework and doubts

Day 1 Homework

1. Print your name.
2. Print your college name.
3. Print your department.
4. Print the result of $10 + 20$.